RUNGTA INTERNATIONAL SKILLS UNIVERSITY

SCHOOL OF INFORMATION TECHNOLOGY

BACHELOR OF COMPUTER APPLICATION (BCA)

PROJECT TEST CASE VERIFICATION REPORT

Project Title: Hospital Patient Data Analyzer (PS No. 17)
Language / Tools: Python 3, Pandas, Matplotlib
Execution File: one.py
Data Source: patients.csv

TEST CASE 1: NORMAL INPUT (DEPARTMENT COUNT & TOTAL PATIENTS)

Objective & Description: Verify department-wise patient count distribution and overall patient volume under normal operating input conditions.

Patient_ID	Age	Gender	Department	Admit_Date	Discharge_Date
P1	45	M	Cardiology	2026-08-01	2026-08-05
P2	30	F	Ortho	2026-08-02	2026-08-04
P3	60	M	Cardiology	2026-08-03	2026-08-08

Expected Output: Cardiology: 2 | Ortho: 1 | Total patients: 3 (Status: PASSED)

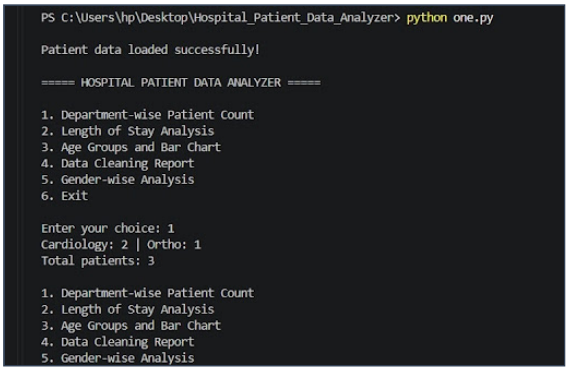


Figure 1.1: Console Output for Menu Option 1 demonstrating department breakdown.

TEST CASE 2: BOUNDARY / MINIMUM INPUT (SAME-DAY DISCHARGE)

Objective & Description: Validate boundary condition where patient is admitted and discharged on the same calendar day (zero days duration).

Patient_ID	Age	Gender	Department	Admit_Date	Discharge_Date
P4	25	F	ENT	2026-08-10	2026-08-10

Expected Output: P4: 0 days | Average stay: 0.00 days (Status: PASSED)

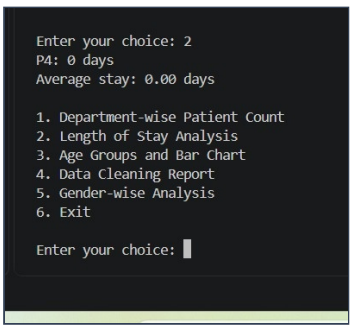


Figure 2.1: Console Output for Menu Option 2 showing calculated 0-day duration.

TEST CASE 3: INVALID INPUT & DATA CLEANING (DUPLICATES & MISSING AGE)

Objective & Description: Test preprocessing logic for duplicate identifier removal and median age imputation on missing values.

Patient_ID	Age	Gender	Department	Admit_Date	Discharge_Date
P1	45	M	Cardiology	2026-08-01	2026-08-05
P2	30	F	Ortho	2026-08-02	2026-08-04
P3	60	M	Cardiology	2026-08-03	2026-08-08
P1	45	M	Cardiology	2026-08-01	2026-08-05
P6	*[Missing]*	M	ENT	2026-08-06	2026-08-07

> **Expected Output:** Duplicates removed: 1 | Missing Age filled with median: 45 (Status: PASSED)

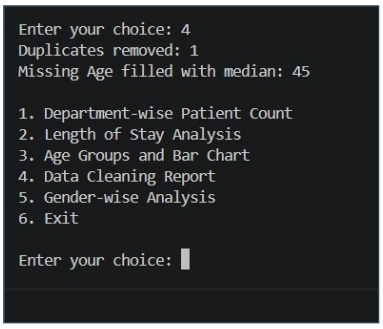


Figure 3.1: Data Cleaning Report verifying duplicate removal and median calculation.

TEST CASE 4: DIFFERENT VALID SCENARIO (STAY & AGE GROUP ANALYSIS)

Objective & Description: Compute mean hospital stay and categorize patients into brackets (0-40, 41-60, 61+) with Matplotlib bar chart visualization.

> **Expected Output:** Average stay: 3.67 days | Age groups: 0-40: 1 | 41-60: 2 | 61+: 0 | Bar chart displayed (Status: PASSED)

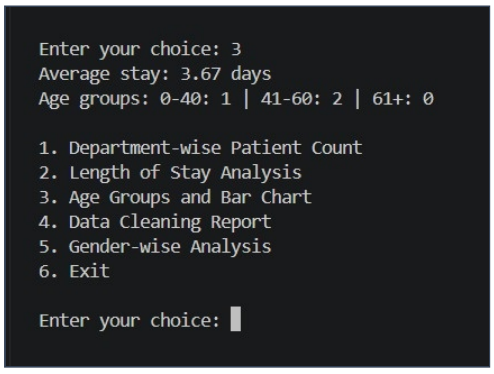


Figure 4.1: Terminal output for stay and age distribution.

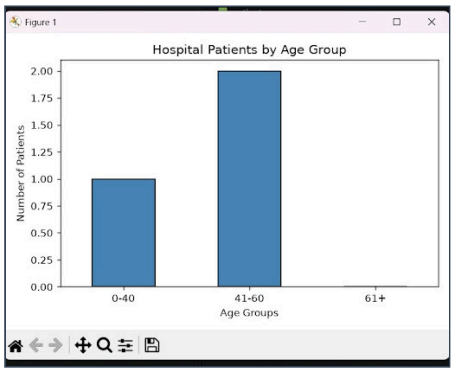


Figure 4.2: Matplotlib bar chart distribution window.